

Q3 SOlution using storyboard drag and drop :

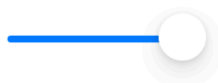
4:35



Count : 0

Increment

switch for Increment button



4:35



Count : -1

Increment

switch for Increment button



4:36



Count : 2

Increment

switch for Increment button



Code: ViewController.h

```
#import <UIKit/UIKit.h>

@interface ViewController : UIViewController

@property (weak, nonatomic) IBOutlet UILabel *countLabel;
@property (weak, nonatomic) IBOutlet UIButton *incrementButton;
@property (weak, nonatomic) IBOutlet UIImageView *imageView;
@property (weak, nonatomic) IBOutlet UISwitch *toggleSwitch;
@property (weak, nonatomic) IBOutlet UISlider *opacitySlider;

@property (assign, nonatomic) NSInteger count;

- (IBAction)incrementButtonTapped:(UIButton *)sender;
- (IBAction)screenTapped:(UITapGestureRecognizer *)sender;
- (IBAction)switchValueChanged:(UISwitch *)sender;
- (IBAction)sliderValueChanged:(UISlider *)sender;

@end
```

ViewController.m

```
#import "ViewController.h"
```

```
@interface ViewController () <UIGestureRecognizerDelegate>
@end
```

```
@implementation ViewController
```

```
- (void)viewDidLoad {
    [super viewDidLoad];
```

```
    self.count = 0;
    [self updateUI];
```

```
    UITapGestureRecognizer *screenTap = [[UITapGestureRecognizer alloc] initWithTarget:self
    action:@selector(screenTapped:)];
    screenTap.delegate = self;
    screenTap.cancelsTouchesInView = NO;
    [self.view addGestureRecognizer:screenTap];
```

```
    [self.toggleSwitch setOn:YES animated:NO];
    self.incrementButton.enabled = YES;
```

```

    self.opacitySlider.minimumValue = 0.0;
    self.opacitySlider.maximumValue = 1.0;
    self.opacitySlider.value = 1.0;
}

- (IBAction)sliderValueChanged:(UISlider *)sender {
    self.imageView.alpha = sender.value;
}

- (IBAction)switchValueChanged:(UISwitch *)sender {
    self.incrementButton.enabled = sender.isOn;
    self.incrementButton.alpha = sender.isOn ? 1.0 : 0.5;
}

- (IBAction)incrementButtonTapped:(UIButton *)sender {
    self.count++;
    [self updateUI];
}

- (IBAction)screenTapped:(UITapGestureRecognizer *)sender {
    self.count--;
    [self updateUI];
}

#pragma mark - UIGestureRecognizerDelegate

- (BOOL)gestureRecognizer:(UIGestureRecognizer *)gestureRecognizer
shouldReceiveTouch:(UITouch *)touch {
    if (!self.toggleSwitch.isOn) {
        return NO;
    }

    if ([touch.view isDescendantOfView:self.incrementButton]) {
        return NO;
    }
    if ([touch.view isKindOfClass:[UISwitch class]]) {
        return NO;
    }
    if ([touch.view isKindOfClass:[UISlider class]]) {
        return NO;
    }

    return YES;
}

```

```
- (void)updateUI {  
    self.countLabel.text = [NSString stringWithFormat:@"Count : %ld", (long)self.count];  
  
    if (self.count == 0) {  
        self.view.backgroundColor = [UIColor whiteColor];  
    } else if (self.count >= 1) {  
        self.view.backgroundColor = [UIColor greenColor];  
    } else {  
        self.view.backgroundColor = [UIColor redColor];  
    }  
}  
  
@end
```